Pinboard Studio - Project 2 Execution Plan

Issue dependencies, recommended team order, owner split, links, and rubric grading map

How to use this plan

Complete phases in order. Within a phase, work may run in parallel only when the issue has no unmet dependencies shown below. Open each GitHub issue first because it contains the full detailed implementation steps and acceptance criteria. Issues marked Complete only when the issue acceptance criteria are demonstrated.

Current scoring summary in the supplied rubric: the top of the PDF states 335 points possible. Some rubric text is truncated or poorly rendered, especially the video and Vision/Gemini point ranges. Confirm the final Canvas point allocation before submission; the table below maps every readable criterion.

Recommended execution schedule

Phase 1 - Foundation

#2 - Project foundation: replace scaffold with production-ready Express structure

Owner: RyanTorrs | Area: Foundation

Dependencies: None

https://github.com/bhushannakhate/project-2/issues/2

#22 - Configure Firebase project, environment variables, and service account access

Owner: RyanTorrs | Area: Foundation

Dependencies: Project foundation: replace scaffold with production-ready Express structure

https://github.com/bhushannakhate/project-2/issues/22

#14 - Configure CI, linting, formatting, and pull-request checks

Owner: bhushannakhate | Area: Foundation

Dependencies: Project foundation: replace scaffold with production-ready Express structure

https://github.com/bhushannakhate/project-2/issues/14

Phase 2 - Identity and data

#16 - Implement Firebase Authentication API and protected routes

Owner: RyanTorrs | Area: Authentication

Dependencies: Configure Firebase project, environment variables, and service account access

https://github.com/bhushannakhate/project-2/issues/16

#21 - Define Firestore collections, indexes, security rules, and data helpers

Owner: bhushannakhate | Area: Database

Dependencies: Configure Firebase project, environment variables, and service account access

https://github.com/bhushannakhate/project-2/issues/21

#11 - Build the React sign-in and account experience

Owner: bhushannakhate | Area: Authentication UI

Dependencies: Configure Firebase project, environment variables, and service account access, Implement Firebase Authentication API and protected routes

https://github.com/bhushannakhate/project-2/issues/11

#4 - Implement authenticated user profile API and credits

Owner: bhushannakhate | Area: Credits

Dependencies: Implement Firebase Authentication API and protected routes, Define Firestore collections, indexes, security rules, and data helpers

https://github.com/bhushannakhate/project-2/issues/4

#17 - Implement creation CRUD API

Owner: bhushannakhate | Area: Database

Dependencies: Implement Firebase Authentication API and protected routes, Define Firestore collections, indexes, security rules, and data helpers

https://github.com/bhushannakhate/project-2/issues/17

Phase 3 - Storage, dashboard, and Pinterest

#3 - Implement Cloud Storage lifecycle for source, output, and audio media

Owner: bhushannakhate | Area: Storage

Dependencies: Project foundation: replace scaffold with production-ready Express structure, Configure Firebase project, environment variables, and service account access

https://github.com/bhushannakhate/project-2/issues/3

#1 - Implement image source upload and validation

Owner: bhushannakhate | Area: Storage

Dependencies: Implement Cloud Storage lifecycle for source, output, and audio media

https://github.com/bhushannakhate/project-2/issues/1

#18 - Build responsive dashboard and creation library

Owner: RyanTorrs | Area: Frontend

Dependencies: Implement creation CRUD API, Build the React sign-in and account experience

https://github.com/bhushannakhate/project-2/issues/18

#15 - Implement Pinterest OAuth 2.0 connection and user token storage

Owner: bhushannakhate | Area: Pinterest

Dependencies: Configure Firebase project, environment variables, and service account access, Implement Firebase Authentication API and protected routes

https://github.com/bhushannakhate/project-2/issues/15

#19 - Build Pinterest boards and pin browser with image import

Owner: bhushannakhate | Area: Pinterest

Dependencies: Implement Pinterest OAuth 2.0 connection and user token storage, Implement image source upload and validation, Implement Cloud Storage lifecycle for source, output, and audio media

https://github.com/bhushannakhate/project-2/issues/19

Phase 4 - AI editing workflow

#5 - Create text and optional voice instruction editor

Owner: RyanTorrs | Area: Editing UI

Dependencies: Implement image source upload and validation, Implement Cloud Storage lifecycle for source, output, and audio media

https://github.com/bhushannakhate/project-2/issues/5

#24 - Implement Gemini Vision analysis and editing prompt preparation

Owner: RyanTorrs | Area: Gemini/Vision

Dependencies: Configure Firebase project, environment variables, and service account access, Implement image source upload and validation

https://github.com/bhushannakhate/project-2/issues/24

#12 - Build audio transcription endpoint and Cloud Storage integration

Owner: RyanTorrs | Area: Audio/AI

Dependencies: Create text and optional voice instruction editor, Implement Cloud Storage lifecycle for source, output, and audio media

https://github.com/bhushannakhate/project-2/issues/12

#8 - Implement Gemini image generation workflow with credits

Owner: RyanTorrs | Area: Gemini

Dependencies: Implement Gemini Vision analysis and editing prompt preparation, Create text and optional voice instruction editor, Implement authenticated user profile API and credits, Implement Cloud Storage lifecycle for source, output, and audio media

https://github.com/bhushannakhate/project-2/issues/8

#10 - Implement review and one-revision workflow

Owner: RyanTorrs | Area: Review

Dependencies: Implement Gemini image generation workflow with credits, Implement creation CRUD API

https://github.com/bhushannakhate/project-2/issues/10

#23 - Create multi-agent orchestration with Google ADK

Owner: bhushannakhate | Area: Agents

Dependencies: Implement Gemini Vision analysis and editing prompt preparation, Create text and optional voice instruction editor, Build audio transcription endpoint and Cloud Storage integration, Implement Gemini image generation workflow with credits, Implement review and one-revision workflow

https://github.com/bhushannakhate/project-2/issues/23

#26 - Publish approved creations to Pinterest

Owner: RyanTorrs | Area: Publishing

Dependencies: Implement Pinterest OAuth 2.0 connection and user token storage, Implement review and one-revision workflow, Implement Cloud Storage lifecycle for source, output, and audio media

https://github.com/bhushannakhate/project-2/issues/26

Phase 5 - Quality, operations, and release

#20 - Add automated API, integration, and frontend tests

Owner: RyanTorrs | Area: Testing

Dependencies: Project foundation: replace scaffold with production-ready Express structure, Implement creation CRUD API, Build the React sign-in and account experience

https://github.com/bhushannakhate/project-2/issues/20

#25 - Deploy backend and frontend to Google Cloud with public URLs

Owner: bhushannakhate | Area: Deployment

Dependencies: Implement Gemini image generation workflow with credits, Implement review and one-revision workflow, Publish approved creations to Pinterest, Configure CI, linting, formatting, and pull-request checks

https://github.com/bhushannakhate/project-2/issues/25

#6 - Add structured logging, metrics, alerts, and Cloud Run evidence

Owner: RyanTorrs | Area: Operations

Dependencies: Deploy backend and frontend to Google Cloud with public URLs

https://github.com/bhushannakhate/project-2/issues/6

#7 - End-to-end usability, accessibility, and responsive QA pass

Owner: bhushannakhate | Area: QA

Dependencies: Deploy backend and frontend to Google Cloud with public URLs

https://github.com/bhushannakhate/project-2/issues/7

#9 - Complete README, architecture documentation, setup guide, and GitHub Wiki

Owner: bhushannakhate | Area: Documentation

Dependencies: Project foundation: replace scaffold with production-ready Express structure, Deploy backend and frontend to Google Cloud with public URLs

https://github.com/bhushannakhate/project-2/issues/9

#13 - Record rubric evidence: screenshots, database CRUD, commits, issues, and demo video

Owner: bhushannakhate | Area: Evidence

Dependencies: Deploy backend and frontend to Google Cloud with public URLs, Add structured logging, metrics, alerts, and Cloud Run evidence, Complete README, architecture documentation, setup guide, and GitHub Wiki

https://github.com/bhushannakhate/project-2/issues/13

Stepwise handoff plan

Bhushan should start: #2 foundation, then coordinate with Ryan on Firebase while Bhushan completes storage, data/credits, and frontend integration.
Ryan should start: #22 Firebase configuration only after #2, then #16 Authentication and later #24 Vision.
Both should hold parallel frontend/backend integration meetings when each phase changes handoff status. Generation (#8) should not begin until image import (#1), credit profile (#4), editing instructions (#5), storage (#3), and Vision (#24) are ready.

Sprint	Handoff	Issues	Division of work
1	Foundation and CI	#2 and #14	Ryan completes backend foundation; Bhushan completes CI/infrastructure conventions.
2	Firebase/data contract	#22, #21, #16	Ryan configures Firebase/auth; Bhushan defines Firestore and protected APIs.
3	User and CRUD vertical slice	#11, #4, #17	Bhushan owns UI profile/data integration; Ryan validates auth API behavior.
4	Media and Pinterest slice	#3, #1, #15, #19	Bhushan owns storage/OAuth/import; Ryan supplies API contracts and reviews failures.
5	AI workflow	#5, #12, #24, #8, #10, #23, #26	Ryan owns input/transcription/Generation/Review; Bhushan owns orchestration and publishing integration.
6	Release	#20, #25, #6, #7, #9, #13	Bhushan owns deployment/docs/evidence; Ryan owns tests/logging/QA support.

Rubric grading map and evidence checklist

Rubric criterion	Points	What earns full credit	Issue evidence
Documentation - application intro	5	Purpose, application name, public deployed URL and team context.	README, issue 9, and issue 13
Development description	5	Commit history, issue-board snapshots over time, major-file descriptions, development narrative.	Issues 13 and 14
Demonstration of application	10	Separate evidence for both users and Pinterest image import plus resulting application behavior.	Issues 13 and 19
Google Database/Firestore	20	Setup, structure, CRUD from app, queries, before/after data and retrieval evidence.	Issues 21, 17, and 13
Code styling and comments	20	Consistent CamelCase naming and thorough source comments, as required by rubric.	Issues 2 and 14
Code repository completeness	5	All code, HTML, images, media and supporting files in GitHub.	All implementation issues
Analytics frontend and backend	15	Both Google Analytics integrations, useful trends and working implementation.	Issue 6 and issue 13
Analytics visuals and clarity	5	Clear charts/visuals suitable for explaining results.	Issue 13
Analytics interpretation	5	Comprehensive explanation of trends and what they mean for users/the app.	Issue 13
Analytics limitations	5	Accurate and detailed discussion of data, analysis, and conclusion limitations.	Issue 13
Cloud logging	10	Structured implementation and demonstration of calls/APIs in Cloud Logging.	Issue 6
Interface functionality	10	Secure import, task selection, loading/success/error feedback, seamless core workflows.	Issues 5, 18, 19, 10, and 7
Interface style	10	Visual appeal, consistency, responsive layout and accessibility.	Issue 7
Fully functioning application	60	Reliability, completeness, external services, Pinterest and Gemini/Vision integration.	Issues 8, 10, 19, 26, 25, and 7
Firestore integration	40	Secure stored image metadata/results, unique IDs/users, retrieval and CRUD.	Issues 21, 17, 3, and 8
Vision and Gemini integration	points vary	Meaningful Vision advanced feature, Gemini prompting/results, error handling and documentation.	Issues 24 and 8
Google Cloud deployment	40	Public Cloud Run/GAE deployment with stable startup, secure configuration, scaling and working features.	Issue 25
YouTube video and final summary	30 stated or source-ordering	Overview, functional requirements, GitHub, app demo, PaaS console, database, and summary sections.	Issue 13